

Structural aliasing in Chronos-Bolt

The five Bayesian models of Deliverable 2: implementation, fitting procedure, gates and verdicts

PATCHALIAS · final implementation · companion report to
chronos/bayesian/notebooks/final_implementation/deliverable2_bayesian_models.ipynb

This report documents the notebook that fits the five model families of Deliverable 2 to the fifteen retrained patch/stride geometries. It states what the notebook implements, which choices the submitted documents already fix and which were left open, what was chosen for each open one and why, how the fitting proceeds, what every gate is and where its threshold comes from, and what the notebook cannot establish. It is written to be read before the notebook is run and beside its output afterwards.

1 What the notebook implements, and what decides what

Deliverable 2 specifies five models, one estimand each, and a decision rule per claim. Deliverable 3's Appendix B states the same five models in full, with the parameterisation, the sampler settings and the reading rule for model comparisons. Where the two documents differ the appendix is followed, for two reasons: it is the same specification written out rather than a new one, and it carries the correction to the one sign error the earlier table contains.

Claim	Model	Estimand	Rule	Response fitted
H1 behavioural	A	$\exp(\beta)$	$\Pr(\beta < \log 0.8) \geq 0.95$; refuted at $\Pr(\beta < \log 1.1) \geq 0.95$	paired recovery contrast d, Eq. (8)
M1 mitigation	A'	δ_O	$\Pr(\delta_O > 0) \geq 0.95$ and the overlap term wins LOO	the same contrast, read across geometries
H1 representational	B	$\exp(\theta_{\text{lock}})$	$\Pr(\theta_{\text{lock}} > \log 1.2) \geq 0.95$; refuted at $\Pr(\theta_{\text{lock}} < \log 1.1) \geq 0.95$	prequential probe codelength in bits
H2	C	σ_ϕ	$\Pr(\sigma_\phi < \log 1.1) \geq 0.95$	the per-phase deficit $y = -d$
H3 location	D1	θ_S, θ_P	the two-label fit wins LOO and both are negative	log token collapse on the union grid
H3a, H3b	D2	κ_S, κ_P	$\Pr(\kappa_F - 1 < 0.1) \geq 0.95$, after an identification gate of ten unambiguous sites	detected site spacing

Table 1. The claims as the notebook implements them. Every threshold is the one `tab:bayesDecisions` preregisters; none was chosen after a posterior was inspected.

1.1 The erratum on the sign of the mitigation slope

Deliverable 2 prints the mitigation rule as $\Pr(\delta_O < 0 \mid D) \geq 0.95$. The response of Eq. (8) is a contrast that rises towards zero as recovery at a candidate frequency approaches recovery at its controls, so a geometry in which the loss is smaller has a **higher** contrast, and an overlap that mitigates therefore produces a **positive** slope. Deliverable 3 and its Appendix B give the rule as $\Pr(\delta_O > 0 \mid D) \geq 0.95$ and state that reasoning explicitly. The positive sign is implemented. The correction is carried in the notebook's own decision table and printed as a boxed note when that table is built, rather than applied silently.

1.2 The design is checked against the documents before anything is measured

Three properties of the submitted report are recomputed from the arithmetic at the top of the notebook, so that a divergence between this implementation and the document appears before any observation exists. Each one stops the notebook if it fails.

- **Fifteen geometries, every stride dividing the 480-sample context.** This is the property that made p32-s28 ineligible: its stride does not divide the analysis context, so it can be swept but not fitted.
- **The exclusive-site counts of `tab:branchSites`.** The notebook recomputes, per geometry, the frequencies one grid predicts and the other does not, and requires the totals to be 36 stride-only against 102 patch-only. Those 36 are the entire identification budget of the stride branch, so reproducing them is not a formality.

- **The correlation of the two centred configuration covariates.** Appendix B reports 0.43 and uses it to argue that the overlap slope and the patch-size slope are separately estimable. The notebook recomputes it and refuses to continue if it differs by more than 0.01, which would mean the centring or the population is not the one the appendix describes.
- **The prior probability of every decision rule.** tab:bayesDecisions publishes, beside each rule, the probability that rule already scores under the priors. Those figures are recomputed analytically from the implemented priors and compared with the published ones. This is the single strongest check that the priors being fitted are the priors that were preregistered, and a mismatch stops the run.

2 The models as implemented

2.1 Models A and A', the behavioural reading and the mitigation

$$d_i \sim \text{Student-}t_4(\mu_i, \sigma), \quad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]}, \quad \beta_c \sim N(\beta + \delta_O \tilde{O}_c + \delta_P \log P_c, \tau)$$

One index per grouping the observation belongs to: c for the configuration, k for the lock harmonic, b for the background realisation. $\tilde{O}_c$ is the centred overlap ratio and $\log P_c$ the centred log patch size, both averaged over the fifteen runs, which is what leaves β the baseline at the average configuration of the design rather than an extrapolation to zero overlap and a patch of one sample. The documents write that intercept with a bar over it; this report writes it β , and the notebook calls it beta_bar.

The response is a paired difference rather than a level, so the effect of interest is the intercept and no lock indicator is needed: the differencing happens in the response and that is why β carries H1 directly. The likelihood is Student- t_4 rather than Gaussian because an arm whose forecast rebuilds nothing sends the logarithm of its recovery ratio far negative; a Gaussian charges such a residual quadratically and a handful of them would set the estimate, while the heavy tail admits them without obeying them. That is what lets every collected row stay in the fit, including the arms where the predicted loss should be largest.

The background index is the pair (generator, realisation) rather than the realisation number alone. The two generators draw their realisation zero independently, so indexing on the number would place two unrelated signals in one level, and the scale σ_b , which Appendix B says is what separates generator-specific variation from the phenomenon, would be measuring something else.

2.2 The parameterisation Appendix B requires, and what it fixes

Appendix B states two rules that no earlier notebook in this project implemented. Both are implemented here, and the first is the reason the earlier fits failed.

Sum to zero. The appendix's words: because the offsets enter the same linear predictor additively, a constant added to one set and subtracted from another leaves the likelihood unchanged, so each set is constrained to sum to zero; without that constraint the levels are identified only through their priors and the sampler is left to explore a ridge along which they trade off exactly. The earlier runs of this project reported R-hat up to 2.23 at an effective sample size of 5 **with zero divergences**, which is the signature of exactly that ridge. A funnel announces itself through divergences; a ridge is a direction along which parameters compensate, which the sampler crosses slowly without ever failing a step, so no amount of re-parameterising a funnel could have fixed it.

The constraint is applied to the **offset** sets: the harmonic and background offsets of Models A and C, the phase offsets of Model C, and the probe-stage and geometry offsets of Model B. It is deliberately **not** applied to β_c . β_c is not an offset around a separate intercept: it is the configuration level itself, and it carries β , δ_O and δ_P through its prior mean. Constraining it to sum to zero would delete the regression M1 is read from. With the offset sets constrained the additive constant has nowhere left to go, which is what identifies the model, and that reading of the appendix is stated in the notebook where the builder is defined.

Centred or non-centred, per group. The appendix chooses the form per group rather than once per model: the configuration, harmonic and background effects are informed by hundreds to thousands of observations each and are written centred, while the eight phase offsets are written non-centred because H2 predicts their scale near zero, which is the regime the non-centred form exists for. Both are implemented as written, so the phase offsets are $u_p = \sigma_\phi z_p$ with z_p standard normal **and** sum-to-zero.

One consequence is stated rather than glossed. A sum-to-zero normal of length n and scale σ has component standard deviation $\sigma\sqrt{(1-1/n)}$, so σ_ϕ is the scale parameter of the constrained distribution and not the realised spread of the eight offsets. The decision rule is written on the parameter and is applied to the parameter; the realised spread is reported alongside as a separate quantity so the two cannot be confused.

2.3 Models B, C, D1 and D2

$$L_i \sim \text{Gamma}(r, r/\mu_i), \quad \log \mu_i = \alpha_0 + \theta_{\text{lock}} \text{IsLocked}_i + s_{\text{st}[i]} + g_{\text{geo}[i]}$$

A codelength is strictly positive, right-skewed, and has a variance that grows with its mean, which is what the Gamma likelihood with a log link is for; the log link is what makes the estimand multiplicative. The stage and geometry offsets are zero-mean and absorb level differences that run to several-fold between probe stages and would otherwise be available to the lock indicator. The unadjusted form, Eq. (10) taken literally, is fitted alongside and compared by leave-one-out, so the blocking decision is reported rather than assumed.

$$y_i \sim \text{Student-}t_4(\mu_i, \sigma), \quad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]} + u_{p[i]}, \quad u_p \sim N(0, \sigma_\phi)$$

Model C is Model A on the per-phase deficit with the configuration covariates dropped and the phase circle cut into eight equal slices. Slices are preferred to a fitted sinusoid because they presuppose no shape: any systematic pattern around the circle, of whatever form, widens the spread. Two readings are reported and they can disagree: the magnitude reading $\Pr(\sigma_\phi < \log 1.1)$, and the evidential reading, a LOO comparison against the same model with the phase term removed. Only the first is the rule; the second is reported beside the verdict and does not gate it, which is what `tab:bayesDecisions` specifies. Appendix B adds a condition that is easy to miss: where the population effect is itself indistinguishable from zero, little remains for phase to move, so H2 is read conditionally, and the deficit scale of the same fit is printed for that purpose.

$$\log z_g(f) \sim N(\alpha_g + \theta_s \cdot \text{OnStrideGrid} + \theta_p \cdot \text{OnPatchGrid}, \sigma)$$

Four labellings of the same profiles are fitted and compared: both grids, stride only, patch only, neither. H3 predicts $\theta_s < 0$ and $\theta_p < 0$, so the support probability the notebook computes is the probability of the **joint** event and not of two marginals each above the cutoff. Two marginals at 0.96 can accompany a joint probability well below it when the two coefficients are negatively correlated in the posterior, and the hypothesis is the conjunction. Each generator is fitted separately and the verdict takes the less favourable of the two, because a location law that held on one corpus and not the other would not be a property of the tokenisation.

$$f_{1,g}^F \sim N(\kappa_F \Delta_g^F, \sqrt{(\sigma_F^2 + \Delta_f^2)}), \quad \Delta_g^S = f_s/S_g, \quad \Delta_g^P = f_s/P_g$$

The residual carries the floor Δ_f the sweep step, because a spacing read off a discrete grid is no more precise than one step. Without it, sites landing exactly on the prediction drive σ_F to zero and the posterior develops a funnel the sampler cannot traverse; with it, a near-zero σ_F is a result rather than a sampler failure. Both responses of Appendix B are fitted: the fundamental, which is the specified one, and the median gap between successive sites as a robustness check.

3 The choices, and the reason for each

Some of what a fitting notebook must decide is already fixed by the submitted documents; the rest was open. The distinction matters, so the two are separated here, and every threshold the notebook applies is labelled in its own output as either preregistered or operational.

3.1 Fixed by Appendix B, and therefore not a choice

Setting	Value	Why it is not free
Chains	4	Appendix B, sampling and diagnostics.
Warm-up draws	2000	Appendix B. Used to adapt the step size, then discarded.
Retained draws	2000	Appendix B. 8000 posterior draws in total.
Target acceptance	0.9	Appendix B: a shorter step for a more cautious traversal of the posterior geometry.

Setting	Value	Why it is not free
Applied to	every fit	Appendix B's stated reason for fixing them is that no model may receive a more or less careful exploration than another. The auxiliary fits, recovery and the sensitivity ladder included, therefore use the same settings.
Diagnostics	R-hat < 1.01, bulk and tail ESS > 1000, zero divergences	Appendix B, preregistered and gating interpretation.
LOO reading rule	inconclusive below twice the standard error of the difference	Appendix B, so that a nominal ranking is not mistaken for evidence.
Sensitivity ladder	×0.5, ×1, ×2 on every prior scale	Deliverable 3 §4: every effect prior is centred on no effect and its scale is varied over {0.25, 0.5, 1.0}. Expressed as a factor it generalises to the families whose tabulated scale is not 0.5, which is how the notebook applies it.
Phase slices	8	Eq. (11).
Contrast stabiliser	+0.01	Eq. (8).
D2 identification bar	10 unambiguous sites per branch	tab:bayesDecisions, applied before either branch rule is read.

Table 2. The notebook prints these at the top of every run, so the record says what was used rather than what was intended.

3.2 What the amplitude recovery is measured against

Appendix B defines R as the ratio of the amplitude the forecast places at the frequency the arm carries to the amplitude **injected** there. The estimator in the shared probing library returns something else, and always has: the least-squares amplitude of the **true continuation** at that frequency, which is the injected tone plus whatever the background contributes there. The two are not the same quantity. A forecast that reproduces the tone exactly and none of the background scores 1 under the first and less than 1 under the second, and the difference is not small.

Neither is obviously the wrong measurement, which is why the choice is made explicit rather than settled here. The injected amplitude is one constant shared by all three arms of a triplet, so it cancels from the contrast entirely and d becomes a comparison of forecast amplitudes alone. The continuation's amplitude differs between the three arms, so dividing by it also controls for the background's differing energy at the three frequencies, at the price of carrying that background's noise into the response.

The collection records `a_pred`, `a_true` and `amp_injected` for every arm, so both ratios are formed from one set of forward passes and the choice is not fixed at collection time. `RECOVERY_DENOMINATOR` selects the one that is fitted, it enters the run fingerprint, and it **defaults to the injected amplitude**, which is what the document says. The other is computed alongside, and Part 2.4 prints what the choice does to the response before any model sees it, so the size of the difference is reported rather than assumed negligible. This matters for A, C and M1, which are all read off the same contrast.

3.3 The tone amplitude: the one design constant the report does not fix

Neither deliverable states the amplitude at which the injected tone rides on the unit-variance background, so it is a genuine choice. It is **1.5**, set explicitly in the notebook's control panel rather than inherited from whatever the shared library happens to default to, and it enters the run fingerprint.

It used to need a guard of its own. The amplitude lived in a module variable of the shared probing library rather than in the collection's configuration object, so it did **not** enter that library's design fingerprint, and two collections differing only in amplitude were indistinguishable to the collection manifest. It is now a field of `collect.Config`, which puts it inside the fingerprint, so a directory collected at one amplitude refuses a run that asks for another. §8.1 states that change and the two others made alongside it.

The trade-off is worth recording. A high amplitude makes the tone dominate its background, so recovery sits near one nearly everywhere and a real deficit has little room to show; a low one makes rebuilding the tone a harder task, which gives a candidate-specific deficit more room to appear at the cost of a noisier recovery ratio. The pairing absorbs part of that noise, since the three arms of a triplet share their background realisation and their phase, so only the frequency changes.

3.4 Leave-one-out on the whole design rather than on a subsample

Models A and C carry 371,000 observations. Their log likelihood at four chains and 2000 retained draws is 23.7 GiB, against the 10 GB a hosted runtime has, which is why earlier attempts either crashed or restricted the comparison to a declared subsample of twenty thousand rows. Neither is necessary. Leave-one-out with Pareto-smoothed importance sampling is computed independently per observation, so the array never has to be resident: the notebook rebuilds the log likelihood from the posterior in slices sized to a memory budget and keeps only the three numbers per observation that the calculation needs.

The consequence is that `elpd_loo` here is the expected log pointwise predictive density of the fitted population, not of a subset of it, and the paired standard error of a difference is computed over every observation. What makes that safe rather than merely convenient is §9: both the reconstruction and the ported smoothing are proved against PyMC and ArviZ before anything depends on either.

3.5 The grid labelling of Model D1

Appendix B says a frequency counts as on a grid when its distance to the nearest member of that comb is within tolerance, and names no number. The sweep grid is the union of a uniform one-hertz grid with the **exact** predicted sites of every geometry, so exact membership is available, and the notebook requires it, to 10^{-6} Hz for the rounding a Parquet round trip introduces.

A one-hertz tolerance would be the obvious alternative and it is the wrong one: it also labels the integer neighbours of a non-integer site, 42 and 43 Hz for the 42.67 Hz stride site of $S = 12$, which lie on no comb at all. Those rows are off-grid observations labelled on-grid, and they pull both coefficients towards zero. The one-hertz labelling is fitted as a declared robustness reading rather than discarded, so the size of that attenuation is reported instead of asserted.

Each collapse curve is normalised by its own off-grid median, as the appendix states. Because α_g is a free level per geometry, θ_s and θ_p are invariant to any per-geometry constant rescaling; the normalisation matters only in that it also removes differences between replicates, which α_g does not absorb.

3.6 Parameter recovery: two scenarios, and what the second one is

Deliverable 2 lists parameter recovery among its validation steps and says what it is for: data simulated with a known effect, to confirm the model finds it, which is what separates *no effect* from *a design that cannot see one*. Two scenarios are run per family, because one answers only half of that. The first simulates the effect the claim is about and asks whether the design can see it; the second simulates the state the claim denies and asks whether the model returns a posterior on that instead, which is what a reported null has to rest on.

What the second state is depends on the claim and it is not always zero. For A, B, C and D1 it is no effect. For D2 it is a slope that is **not one**: H3a and H3b claim that a branch tracks its own parameter, so their negation is that it tracks it imperfectly, and the scenario is $\kappa_F = 0.65$. The question there is whether the posterior can tell 0.65 from 1.

Coverage alone would be too weak a gate to say what a pass here is taken to say. An interval wide enough to contain both scenarios' truths covers each of them and demonstrates nothing: it shows that the estimator is not biased, not that the design can tell an effect from its absence. The gate therefore has a second half. Each scenario's interval must also **exclude the other scenario's truth**, in at least the same share of repeats. Coverage says the estimate is in the right place; discrimination says the design can resolve the two places apart. The mean interval width is reported beside the mean separation between the two truths, which is the comparison that makes a marginal case legible.

The group offsets are simulated sum-to-zero, matching the prior they are recovered under. An independent draw has a group mean of order $\sigma_u/\sqrt{n}$, which the constrained model has nowhere to put except the intercept, and the result would be a bias in the test rather than in the estimator. Models A and C recover on a declared stratified subsample of twelve thousand rows, because a recovery run at the full 371,000 would cost as much as the reported fit and buy nothing: a model that recovers a truth at reduced n recovers it at full n with a narrower interval.

3.7 Thresholds this notebook declares rather than inherits

Threshold	Value	What it gates, and why this value
Predictive coverage	0.90	The share of the levels of every stratum whose observed statistic must fall inside the 95 per cent replicated interval. A 95 per cent interval covers about 95 per cent of levels when the model is right, so requiring 0.95 would be a coin flip at the nominal rate; 0.90 leaves room for that sampling variation while still failing a model that misses a stratum systematically.
Sensitivity band	0.10	The admissible excursion of a rule probability across the three rungs of the ladder, and additionally the decision class must be identical at every rung. Two converged variants do not excuse a third that failed.
Recovery coverage	0.80	The share of repeats whose 95 per cent interval must contain their own truth, per parameter and scenario.
Recovery discrimination	0.80	The share of repeats whose interval must exclude the other scenario's truth. Without it, an interval wide enough to contain both would pass, and a pass would mean only that the estimator is unbiased.
Recovery subsample	12,000 rows	Stratified, for Models A and C only; B, D1 and D2 recover on their whole tables.
Predictive draws	400	Drawn in batches of 25, so the replicate array is never built in full. At 400 draws a stratum interval is estimated to about half a per cent.
LOO slice budget	192 MiB	The resident size of one observation slice, which sets the slice width from the number of posterior draws.

Table 3. These are operational and the notebook labels them as such in its own output and in the run manifest. None is attributed to the report.

3.8 The environment, and why not the project lock file

The notebook installs a short pinned list and restarts the runtime once, verifying the numpy, scipy, ArviZ and PyMC import chain in a clean subprocess before trusting it. Exporting the project's lock file and installing all of it is what earlier notebooks did, and it is avoided for two reasons those notebooks record in their own comments. A hosted runtime imports numpy into the kernel process before any user cell runs, so installing a different numpy leaves an already-compiled extension resident beside newer Python source and a later ArviZ import fails several frames deep inside scipy with a message that blames neither. And installing a locked torch over the runtime's own build leaves its torchvision behind with a mismatched ABI, so importing the forecasting package dies on a missing operator. Torchvision and torchaudio are not dependencies of this project at all, so they are removed rather than version-matched.

The sampler implementation is a control rather than a constant. The reference is PyMC's own NUTS; a Rust implementation of the same sampler is offered as an option because it is several times faster on the same processor. Both target the same posterior with the same priors, the same number of draws and the same target acceptance rate, so the choice is one of throughput, the random stream differs, and every diagnostic applies unchanged. Whichever is used is recorded in the run fingerprint.

3.9 Where the implementation interprets, corrects or departs from the documents

Every such place, in one list. The notebook prints the same list at the top of every run and saves it as `00_specification_notes.csv`, so a reader who disagrees with a row can find the control that changes it rather than having to discover the decision.

Kind	Subject	What the document says	What is implemented
erratum	the M1 sign	Deliverable 2 prints $\Pr(\delta_O < 0) \geq 0.95$	$\Pr(\delta_O > 0) \geq 0.95$, as Deliverable 3 and Appendix B have it
interpretation	sum-to-zero	Appendix B: each set of offsets is constrained to sum to zero	applied to the offset sets, and not to β_c , which is the configuration level itself and carries the M1 regression
choice	the denominator of R	Appendix B defines R against the amplitude injected; the shared estimator returns the true continuation's amplitude	both collected, the injected one fitted by default, the control in the fingerprint (§3.2)
choice	the D1 grid tolerance	Appendix B: within tolerance, with no number given	exact membership as primary, one hertz as a declared robustness reading (§3.5)

Kind	Subject	What the document says	What is implemented
departure	the D1 signal modes	Appendix B: all three modes are fitted	all three fitted and reported; the verdict is read from the two generator modes, because on a pure sinusoid the response depends on the floor and a stratum of identical values has no dispersion for a predictive check to reproduce
generalisation	the sensitivity ladder	Deliverable 3: every effect prior is varied over {0.25, 0.5, 1.0}	a factor of 0.5, 1 and 2 on every prior scale, which reproduces those three where the tabulated scale is 0.5 and extends the rule to the other families
addition	Model D2's two readings must agree	tab:bayesDecisions gates D2 on ten unambiguous sites only	an operational gate: the preregistered fit and the fit restricted to geometries whose lowest detected site is the fundamental must reach the same decision (§6.1)
addition	operational thresholds	the report names the checks but not their thresholds	the five values of Table 3, declared here and recorded in the run manifest

Table 4. Nothing in this list was decided after a posterior was seen; every row is fixed by the notebook before Part 2 collects anything.

4 The fitting procedure, stage by stage

Part	What happens	What it produces
0	Locate the checkout, install and verify the environment, resolve the run folder, check the design and the priors against the documents, resolve the stage, print the run plan.	<code>analysis_manifest.json</code> , the prior and decision tables, the inventory of what is done and what is still to do.
1	Prior predictive for all five families, on the real design axes, with the offsets drawn under the constrained prior that will be fitted.	Four figures and a summary table, written before any observation exists.
2	Collect the five tables from Chronos, one geometry at a time; derive the triplet contrast; gate the collected design for these five models.	Five Parquet tables with a manifest and per-file hashes, the contrast audit, the phase mapping, the design gate, four raw-data figures.
3	The likelihood written out and checked against SciPy; the five specifications; the chain-blocked sampler; the log-likelihood reconstruction and its proof; streaming PSIS-LOO and its proof; parameter recovery in two scenarios per family.	Two parity tables, the recovery table and its summary, three figures.
4	The fits: Model A and its four configuration levels, Model B adjusted and literal, Model C with and without phase, Model D1 in four labellings per generator plus the tolerance robustness, Model D2 per branch and per response plus the harmonic-order audit.	One checkpoint per fit and per pair of chains, the coefficient tables, the site audit, five figures.
5	Convergence; stratified posterior predictive checks from the reconstruction; the sensitivity ladder on every family; LOO over the whole design; the verdicts.	The convergence, predictive, sensitivity, LOO and verdict tables, the run summary, four figures.
6	Reading the results from the saved artifacts alone: every estimand against its threshold, prior against posterior, chain behaviour, the funnel-risk pairs, Pareto diagnostics, the decision dashboard, the timings and the index.	Ten or more figures and the figure and table indices.

Table 5. Each part is resumable on its own, and a part whose artifacts are already present and hash correctly reloads instead of recomputing.

5 The gates

Interpretation is fail-closed. A gate that fails does not weaken a claim into a hint: the estimand is reported as **not reportable**, and a branch below its site bar as **not identified**, and neither is a null result. The notebook applies seven gates and reports each one as its own column, so a failure is attributable rather than merely noted.

Gate	What it requires	Threshold from	Applies to
Prior calibration	The prior probability of every rule reproduces the Prior column of <code>tab:bayesDecisions</code> , and the prior predictive of every family is finite.	preregistered	the whole run
Design	Every geometry present in every table; both generators and all eight phase slices populated; both levels of the lock indicator in every geometry; the grid labels of full rank against the geometry effects; positive dispersion in the fitted modes.	the models' own identifiability	per model
Reconstruction parity	The rebuilt log likelihood agrees with <code>pm.compute_log_likelihood</code> to better than 10^{-8} , for every family.	this implementation	the whole run
PSIS parity	Every scalar, every per-observation contribution and every Pareto k agrees with <code>az.loo</code> to better than 10^{-6} .	this implementation	the whole run
Convergence	R-hat < 1.01, bulk and tail ESS > 1000, zero divergences, four chains, over every parameter of every fit the claim reads.	preregistered	per claim
Parameter recovery	Both scenarios ran, every fit converged, and at least 80 per cent of repeats cover the truth.	preregistered step, operational threshold	per family
Posterior predictive	The observed mean and standard deviation inside the 95 per cent replicated interval for at least 90 per cent of the levels of every stratum, plus three global quantiles.	preregistered step, operational threshold	per model
Prior sensitivity	All three rungs converged, the rule probability moving by at most 0.10, and the same decision class at every rung.	preregistered step, operational threshold	per claim
LOO reliability	Every Pareto k below ArviZ's threshold for the number of draws, and the required formulation leading by at least twice the paired standard error.	preregistered	M1 and H3 location only , the two claims whose rule has a comparison leg. For H2 and for Model B's blocking the comparison is reported beside the verdict and does not gate it, which is what <code>tab:bayesDecisions</code> specifies.
Identification	For a D2 branch, at least ten unambiguous sites on the replicate-averaged curves of the two generator modes.	preregistered	per branch
D2 readings agree	The preregistered fit and the fit restricted to geometries whose lowest detected site is its branch's fundamental must reach the same decision.	operational, §6.1	H3a and H3b only

Table 6. The first four run once and stop the notebook outright, because a run that fails one of them has produced nothing worth gating. The rest are per claim and appear as columns of the verdict table.

Two properties of that table are easy to get wrong, and both were wrong in an earlier draft of this notebook. A gate that names one representative fit will pass a claim whose other fit failed, so every gate is a **conjunction over every fit the claim is read from**: H3 location pools both generators and takes the less favourable, so a generator whose model fails its predictive check has to fail the gate. And a comparison the rule does not mention must not decide a verdict, so the leave-one-out reading is a gate only where `tab:bayesDecisions` makes it a leg of the rule and is reported without gating otherwise. In the verdict table a gate cell is a pass, a failure, or not applicable; a gate that applies and has no result is a failure, never a pass.

5.1 Why the convergence gate reads every parameter

A population effect can mix perfectly while a group scale does not, and the group scale is part of the model the estimand is conditional on. The notebook therefore takes the maximum R-hat and the minimum effective sample size over every posterior variable of a fit, which for Model A means the fifteen configuration effects, the harmonic and background offsets and the eight scalars, not only β . The variables are iterated rather than stacked into one array,

because a dataset of variables with different dimensions broadcasts into their cartesian product and the minimum would then be read off cells that do not exist.

5.2 Why the paired standard error, and not the difference of two

The comparison quantity is $\sum_i (L_i^1 - L_i^0)$ with standard error $\sqrt{N \text{Var}(L_i^1 - L_i^0)}$, computed on differences taken at the same observation. That is what makes it the error of the difference: two models that agree closely observation by observation have a small standard error on their difference even when each has a large one of its own, and the reverse also happens. A comparison is additionally gated on reliability, because if any Pareto k exceeds its threshold the importance-sampling approximation behind the whole calculation is not trustworthy and the comparison returns inconclusive whatever its margin.

6 The summary table: pre-gate, per-gate, post-gate

The notebook's final table has one row per claim and three readings of each.

- **Pre-gate** is the probability rule applied on its own: what the posterior says, before asking whether the posterior may be believed.
- **Each gate, independently**, as its own column, evaluated on its own. The gates are not combined into a score; they are a conjunction, and a further column names the ones that did not hold.
- **Post-gate** is the reportable verdict, resolved in a fixed order: a non-reportable mode first, then identification, then the gates, then the probability rule, and last the comparison leg for the two claims whose rules include one.

Reporting the first two beside the third is the point. A claim that is supported pre-gate and not reportable post-gate is a claim whose evidence exists but whose conditions were not met, and the gate column says which; that is a different situation from a claim that is inconclusive on the numbers with every gate green, and the two would be indistinguishable if only the final verdict were printed.

6.1 The one gate Model D2 needs and the document does not name

Eq. (13) regresses the **fundamental** of a branch on the spacing that branch predicts. Where the first dip of a comb was not detected, the lowest detected site is a higher harmonic and the ratio sits near an integer above one, so a geometry whose sites follow the grid perfectly can still push κ towards two. Part 4.5 of the notebook reports that harmonic order per geometry and fits a restricted version alongside the preregistered one.

Reporting that audit and then ignoring it in the verdict would be the worst of both, so the two readings are required to **agree**. What that establishes is deliberately modest. Neither reading is an independent verification of the movement law, because the branch assignment already used the predicted grid, and ten unambiguous sites do not change that; the agreement establishes only that the answer does not depend on which of the two sets of sites is read. A claim that survives one and not the other is inconclusive, not resolved by choosing. §10 states the limitation in full.

Verdict	Meaning
SUPPORTED	The support rule reached 0.95 and every applicable gate held.
REFUTED	The refutation rule reached 0.95 and every applicable gate held.
INCONCLUSIVE	The gates held and the rule did not decide. A result in its own right. Also the verdict when a claim reaches its probability but loses the comparison leg of its rule.
NOT REPORTABLE	A gate failed, or the run is not in full mode. Not a null result : the estimand is unavailable, not zero.
NOT IDENTIFIED	The design cannot answer the question, for a reason stated in the row. For a D2 branch that is the site bar, and the design gate reports how many unambiguous sites the branch actually supplied. Not a null result .

Table 7. The same five verdicts appear in the notebook's dashboard, its CSV tables and its run summary, and nowhere are they collapsed into a pass or a fail.

7 Memory, and resuming

The constraint that shaped the implementation is a 10 GB runtime against a Model A design of 371,000 observations. Three things follow from it, and none of them changes a model, a prior or a threshold.

Object	If built naively	What the notebook does
log_likelihood of one Model A fit	23.7 GiB, allocated by PyMC after sampling finishes, when memory is already at its peak	Never built. Sampling runs with it switched off, and the array is rebuilt from the posterior in slices of at most 192 MiB wherever it is needed. The posterior is identical either way, because the log likelihood is computed from the draws and does not influence them.
Posterior predictive replicates	1.1 GiB at 400 draws, 5.9 GiB at the full posterior	Drawn in batches of 25 draws, with only the per-stratum statistics accumulated.
Collection working set	up to 130,000 contexts of 544 samples, twice over for the true continuations, held at once for the largest geometry	Candidate frequencies are processed in blocks, with resident memory reported as it goes.

Table 8. The full arrays are what earlier attempts allocated; the right-hand column is what this notebook does instead.

Resumption is designed at three granularities. The collection shards per table per geometry, so an interrupted session loses only the table it was writing, and a geometry whose shards are all present never loads its model at all. Each fit is drawn in blocks of two chains and each block is checkpointed, so a disconnect costs at most one block rather than a whole fit. And each fit's per-observation leave-one-out contributions are checkpointed on their own, so a comparison that dies part-way resumes at the fit it died on.

Every artifact is written atomically and recorded with its SHA-256 and its byte count. The notebook asks one question about a checkpoint, and it answers by hash rather than by the file's presence: a file with no manifest entry, a manifest entry with no file, and a file whose content has moved under a name already recorded are all refused rather than resumed. The run fingerprint covers the seed, the sampler settings, the tone amplitude, every analysis constant, the hashes of the shared scripts and the hashes of the code objects of the functions that build models and datasets, so editing a cell in place invalidates exactly the checkpoints that depended on it.

8 What the earlier notebooks did differently, and why this one does not

The repository holds twenty Bayesian notebooks. Five differences between them and this one are substantive rather than stylistic, and each is a case where an earlier implementation and the submitted documents disagree.

What an earlier notebook did	What the documents say	What this notebook does
Plain normal priors on the harmonic and background offsets of Models A and C.	Appendix B: each set of offsets is constrained to sum to zero, or the sampler explores a ridge along which the levels trade off exactly.	Sum-to-zero on every offset set, and not on the configuration level, which carries the regression M1 is read from. This is the fix for R-hat 2.23 at an effective sample size of 5 with zero divergences.
Fitted contrasts[contrasts['live']], a response filter applied after collection.	Deliverable 3 §4: no response filter is applied afterwards, so the arms where the forecast rebuilds nothing, which is where the loss should be largest, are kept.	Every complete triplet with finite, non-negative recoveries is fitted. The count the filter would have removed is reported as an audit figure.
1500 warm-up and 1500 retained draws at a target acceptance of 0.95.	Appendix B: 2000 and 2000 at 0.9, for every fit.	The appendix's settings, for every fit including the auxiliary ones.
Two marginal probabilities, each required to exceed the cutoff, for H3's two coefficients.	H3 predicts both coefficients negative, which is a conjunction.	The probability of the joint event, with the two marginals shown beside it because they can both exceed it.
Fitted two of the three swept signal modes.	Appendix B: fitting all three is what shows the conclusion is not an artefact of noise-free inputs.	All three are fitted and reported. The verdict is read from the two generator modes, for the reason in Table 4; that restriction is declared rather than left implicit.
A one-hertz tolerance for the grid labels of Model D1.	Appendix B: within tolerance, with no number given; the sweep grid contains the exact predicted sites.	Exact membership as the primary labelling, with the one-hertz version fitted as a declared robustness reading.

Table 9. No notebook is modified: this one is new and stands on its own. The three changes to the shared scripts are in the next subsection.

8.1 Three changes to the shared collection scripts

Three things the notebook needed were properties of `collect.py` and `probe_lib.py` rather than of any one analysis, and they were fixed there rather than worked around in the notebook. No notebook, old or new, is modified by any of this.

What was wrong	The change	Why in the library and not the notebook
The tone amplitude lived in a module constant, so it was outside the collection's design fingerprint: two collections differing only in it looked identical to the manifest.	<code>Config.tone_snr</code> , threaded through to <code>build_context</code> , which gained an optional <code>amp</code> argument defaulting to the old constant.	The fingerprint is the library's own guarantee. A notebook writing a file beside the data can only detect a mismatch it happens to look for; a field of the configuration is checked by every entry point, including the command line.
<code>collect_contrasts</code> built every context of a geometry before a single forward pass: up to 130,000 contexts of 544 samples, twice over for the true continuations.	Candidate frequencies are forwarded in blocks of <code>Config.sites_per_block</code> , with resident memory reported per block.	The command line has the same problem, and so does every other caller. The blocks partition the candidate list in order and each row's readings depend on that row alone, so the table is unchanged.
<code>check_design</code> gated every collection on criteria written for the hit indicator: a non-constant <code>h</code> , and a per-geometry survival count against the instrument ceiling. Neither is defined for a paired contrast, so a finished collection could be discarded over a criterion that does not apply to it.	<code>check_design</code> , <code>merge</code> , <code>load_collection</code> and <code>collect_all</code> take <code>response</code> , either <code>localisation</code> or <code>contrast</code> . Coverage, schema, both arm levels and both generators are required either way; the rest follow the response. The default is <code>localisation</code> , so every existing caller keeps the behaviour it had.	The alternative was for the notebook to replace the function at run time, which leaves the library still wrong for the next analysis and the substitution invisible to anyone reading <code>collect.py</code> .

Table 10. Two smaller fixes travelled with them: `derive_sites` now records which harmonic of its own comb each branch's lowest detected site is, which is the audit §10 describes; and a candidate frequency whose spectrum yields no peak in band is recorded as a miss rather than failing validation, which previously discarded a geometry's forward passes over a legitimate reading.

One consequence is worth stating plainly. `collect.py` and `probe_lib.py` are hashed into every collection's design fingerprint, so editing them invalidates the manifest of every collection already on disk: an older run directory will refuse to be resumed or merged, and will say so. That is the mechanism working, not failing. It costs nothing here, because this analysis collects into a new folder at an amplitude no previous run used, but an older collection that is still wanted must be re-collected or read outside the manifest.

The changes were verified against the code they replace. With a deterministic stub in place of Chronos, the blocked collector reproduces the original table exactly, row for row and column for column, at block sizes of one, two, three and the whole geometry; the amplitude reaches the signal and changes the readings while leaving every metadata column untouched; `build_context` without the new argument is identical to before; the amplitude changes the design fingerprint and the block size does not; each response applies its own criteria and refuses an unknown one; and a missing spectral peak passes validation while a missing recovery does not.

9 Verifying the implementation itself

Two pieces of this notebook are second implementations of something a library already provides, and two implementations of one formula are two chances to be wrong. Both are therefore proved inside the notebook, before anything is allowed to depend on them, and a disagreement stops the run.

- **The reconstructed log likelihood** is checked against `pm.compute_log_likelihood` on the same posterior and the same rows, for every family and for both of Model A's likelihoods, with agreement required to better than 10^{-8} . The check is cheap, a few hundred rows and a short fit, and what it buys is that every comparison and every predictive check below rests on a formula shown to be the model's own.
- **The ported Pareto smoothing** is checked against `az.loo` on those same fits: the expected log predictive density, its standard error, the effective number of parameters, every per-observation contribution and every Pareto k , to better than 10^{-6} .

The changes to the shared scripts were checked the same way, against the code they replaced rather than against a description of it: §8.1 lists what was compared, and the headline is that the reworked collector reproduces the original

table exactly at every block size.

The gate logic is exercised offline too, because a gate that quietly passes is worse than no gate. The checks that a gate over several fits fails when any one of them fails, that an empty list of fits is a failure rather than a pass, that the comparison decides only the two claims whose rule has a comparison leg, that Model D2's two readings are required to agree and that a disagreement is caught, that an interval wide enough to contain both recovery scenarios fails the discrimination criterion while a narrow one passes both, and that the verdict resolves mode, identification, gates, rule and comparison leg in that order, all run against the notebook's own functions extracted from the file.

The same components were additionally exercised offline against independent references while the notebook was written, on synthetic posteriors rather than on Chronos output. The generalised-Pareto fit, its inverse and the weight smoothing reproduce ArviZ's own functions to 10^{-13} ; each family's log likelihood reproduces the corresponding SciPy density exactly, with the Gamma family agreeing to 3×10^{-14} ; the streaming result reproduces `az.loo` on a hand-built inference object to zero difference on every quantity; slicing the observation axis gives bit-identical results to computing the whole array at once; the replicate simulator reproduces the first two moments of each likelihood; the triplet pivot computes Eq. (8) correctly and refuses a duplicated arm, an unpaired triplet, an unrecognised role and a frequency inconsistent with its role; the grid labelling behaves as designed at both tolerances; and the verdict resolution respects its stated order, including demoting a claim that reaches its probability but loses the comparison leg of its rule.

The notebook was also checked statically: every code cell parses, no cell uses a name an earlier cell has not defined, every timed task is in the declared plan and every planned task is used, every artifact a cell reads is written by an earlier cell, and every cross-reference in the prose points at a section that exists.

What none of that establishes is that the sampler will converge on these data. Nothing can establish that in advance; it is what the convergence gate is for, and it is why the parameterisation of §2.2 matters.

10 Limitations no fit can remove

These belong beside any number taken from the notebook into the report. The notebook prints all of them with its run summary.

- **One training seed per geometry.** A difference between two configurations is a difference between two trained models as well as between two geometries, and the design cannot separate the two. An effect large relative to its posterior spread is unlikely to be an artefact of that choice; an effect near a decision threshold could be.
- **The truncated analysis context.** Of the 480 samples supplied, the patch grid spans only a whole number of strides and the remainder enters no token. The truncation cancels within a triplet, which shares its context length, its background and its phase, but not between geometries, and it is non-zero on exactly the six runs with *S* not dividing *P*, which are the runs that identify the stride branch. The subset carrying that branch is therefore also the subset least like the rest of the design.
- **The token count moves with the stride.** Overlap and sequence length are not independently identified, and lowering the stride also changes the mixture of stride- and patch-derived candidates along the same axis. Neither confound could create a mitigation, but either could inflate one.
- **M1 is a between-model comparison.** The slope is estimated across fifteen independently trained variants over fifteen design points, so it is evidence that the effect varies with geometry in the predicted direction across this grid, and not the effect of changing the overlap of a trained model.
- **Model D2 is conditional on its own selection, and no gate removes that.** Dips are detected from the profile alone, which uses neither *P* nor *S*, but each detected dip is then assigned to the branch whose grid it is near, so regressing a branch's fundamental on that branch's predicted spacing is conditional on a selection that used the same spacing. Ten unambiguous sites do not make it unconditional. The requirement that the two readings agree (§6.1) removes a different and smaller problem, that the answer might depend on undetected fundamentals; it is not a substitute for a measurement of the comb that never touches the predicted grid, and this design does not contain one. A supported H3a or H3b here should be read as: given that dips were detected and fell near this branch's comb, their positions equal the spacing that branch predicts.
- **The stride branch is identified only by pooling.** Nine of the fifteen geometries yield no stride-only frequency at all, and the most any single geometry supplies is eight, below the ten unambiguous sites the identification gate requires. All of this is fixed by the choice of (*P*, *S*) before anything is measured.

- **The tone amplitude is a choice, not a quantity the report fixes.** It is recorded in the collection's design fingerprint, and a run at a different amplitude is refused rather than merged.

11 What the run leaves behind

Location	Contents
analysis_manifest.json	The run specification and one entry per artifact, with its SHA-256 and its byte count.
data/	The five collected tables, their shards, the collection manifest, which records the tone amplitude in its design payload, and the generated-signal archive.
tables/*.csv	Every result table, in the same content as the Parquet the notebook reads: the priors and decision rules, the design gate, the contrast audit, the phase mapping, both parity checks, parameter recovery, the coefficient tables, the site audit, convergence, the predictive checks, the sensitivity ladder, the LOO comparisons and the verdicts.
figures/*.png	Roughly thirty figures across the six parts, indexed at the end of the run.
*.nc	One checkpoint per fit and per pair of chains, posteriors only.
05_run_summary.json	Mode, fingerprint, commit, amplitude, collection hashes, every gate result, every verdict, the measured timings and the limitations above.

Table 11. Keep them together. The verdict table on its own does not carry the conditions that made it readable.

12 Running it

Open the notebook in Colab and choose Run all. The first cell installs the pinned environment and restarts the runtime once, which the platform reports as a crashed session; choose Run all again and it steps past the install. That is the whole procedure: every setting is in one form in Part 0.4, the notebook decides for itself whether it needs to collect observations or only to analyse them, it resumes any stage already finished, and it stops with a named reason if a gate fails.

It is a long run and the notebook is built for that rather than around it. From an empty folder it collects 1,113,000 forecasts before it fits anything and then fits about sixty-five posteriors, eleven of them on the full 371,000 observations. On a processor-only runtime the collection is of the order of ten hours and the sampling is of the same order again. Nothing about that requires supervision, and two levers shorten it: a GPU runtime collects in under an hour, with the analysis afterwards unaffected because it reads the collected tables back from storage; and the faster sampler implementation of §3.7 cuts the sampling substantially while targeting the same posterior.

To do this	Set this
The reportable run	MODE = "full", every other default unchanged.
Rehearse the whole chain in minutes	MODE = "smoke". Every grid shrinks and every verdict is non-reportable by construction.
Check the design, the priors and every gate's plumbing without sampling	MODE = "preflight". It stops at the start of Part 4 with a message saying so.
Regenerate the figures alone, in a fresh runtime, with no Chronos and no sampling	STANDALONE_FIGURES = True. Collection and sampling are refused, every stage reloads from its checkpoint, and a missing artifact is reported by name.
Split the fits across sessions	Turn off the FIT_MODEL_* flags this session should not fit. A model already finished in any session reloads from its checkpoint regardless.
Use a different tone amplitude	Change TONE_SNR and RUN_FOLDER. A run folder written at one amplitude refuses another.

Table 12. Nothing outside Part 0.4 needs editing.

A last point about reading the output. The notebook is deliberately willing to stop. It stops if its priors do not reproduce the published prior probabilities, if the design cannot identify a model, if its own log-likelihood reconstruction disagrees with PyMC, if a checkpoint's hash has moved, or if the collection is incomplete. Each of those messages names what failed and what to do about it. A run that reaches the verdict table has passed all of them, and a verdict of

NOT REPORTABLE in that table is a statement about a gate rather than about Chronos.